A3-PS027-TC190

PS-027: IPv6 Website and Dependency Readiness Auditor

Multi-Stage Protocol Failure Isolation, Runtime Dependency Trees & Happy Eyeballs Decoupled Probing

Team A3-PS027-TC190

AIORI 3.0 National Hackathon • Category: Internet Measurement
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Abstract & Problem Formulation

The Apex Domain Fallacy in Modern Web Architectures

- **The Core Problem:** An apex domain resolving to an AAAA record does *not* make a website IPv6-ready. Modern web applications load dozens of external dependencies (scripts, styles, fonts, CDN images, APIs, analytics, tags) that frequently remain IPv4-only.
- **IPv6-Only Environment Impact:** On IPv6-only networks (5G cellular networks with IPv6-only APNs, pure v6 enterprise/cloud environments), IPv4-only dependencies fail to load, resulting in broken layouts, blocked script execution, and functional collapse.
- **The Happy Eyeballs Masking Effect:** Under dual-stack connections, RFC 8305 silently falls back to IPv4 after a 250ms–300ms delay. This masks broken IPv6 dependencies from site operators while degrading real-world latency.
- **Project Scope:** Inspired by the IETF webres6 initiative, this project implements a headless browser auditing system that maps full runtime dependency trees, executes isolated 4-stage protocol failure diagnostics (DNS, TCP, TLS, HTTP), validates a controlled testbed seeded with IPv4 assets (100% detection rate), and evaluates 30 real-world websites.

Background & Industry Context

The Gap Between Client Adoption and Dependency Readiness

• **India's IPv6 Phenomenon:** India leads the world in subscriber IPv6 adoption (>80% mobile penetration led by major telcos). However, domestic web server and sub-resource adoption lags drastically behind client readiness.

• **Composite Web Architecture:** Modern websites are distributed multi-origin applications. An e-commerce or government portal may load assets from 15+ third-party domains (CDNs, ad networks, payment gateways, authentication providers).

• **Why DNS Queries Alone Are Insufficient:** Querying DNS records does not verify end-to-end transport reachability (firewalls, routing blackholes, TLS SNI mismatches, or HTTP 5xx reverse-proxy errors).

• **Need for Browser Runtime Observation:** Dynamically injected scripts, font-face rules, XHR/fetch calls, and iframe contents can only be discovered by executing JavaScript in a genuine browser DOM context.

Literature & Standards Review

Relevant Standards, IETF Initiatives & Existing Tools

Standard / Tool	Focus Area	Key Insight & Critical Limitation
RFC 8200 (IETF Standard)	IPv6 Base Protocol Specification	Defines IPv6 packet headers and addressing architecture; foundational standard.
RFC 8305 (Happy Eyeballs v2)	Dual-Stack Concurrency Algorithm	Prioritizes IPv6 with a 250ms fallback to IPv4. Crucially, this masks IPv6 dependency failures from measurement tools.
SAP webres6 (IETF v6ops)	Selenium-based IPv6 compliance checker	Pioneering tool, but suffers from Happy Eyeballs bias in Chromium, serial scanning, and lacks protocol stage isolation.
W3C Resource Timing	DOM Network Performance API	Standardizes performance entries, initiator types, and network waterfall timestamps.

Proposed Solution: System Architecture

Multi-Stage Protocol Diagnostics with Real Browser Instrumentation

• **Headless Browser Engine (Chromium/CDP):** Renders full pages, intercepts network events, captures dynamic imports, and records causal initiator hierarchy (HTML parser vs. script vs. fetch).

• **Happy-Eyeballs Decoupled Probing:** Bypasses browser fallback by issuing raw parallel socket probes over pure IPv4 and pure IPv6 (AF_INET6), providing ground-truth measurement.

• **4-Stage Protocol Gauntlet:** Evaluates each unique discovered asset across 4 strict sequential OSI layers:

1. **DNS Stage:** A and AAAA resolution, NODATA, CNAME chains, NXDOMAIN, SERVFAIL detection.
2. **TCP Stage:** Raw IPv6 socket handshake (port 80/443); captures Timeout, ConnRefused, and No Route.
3. **TLS Stage:** SSL handshake over IPv6 with SNI; detects expired certs, untrusted CAs, and hostname mismatches.
4. **HTTP Stage:** Strict IPv6 HTTP/1.1 & HTTP/2 retrieval; measures TTFB and flags HTTP 4xx/5xx status codes.

Solution Optimization & Innovation

Key Differentiators & Technical Optimizations

Dimension	Traditional / Existing Approach	Our Proposed Optimization
Inspection Scope	Apex domain DNS AAAA record check only	**Full Browser DOM Execution:** Captures 100% of runtime JS, CSS, fonts, and third-party API dependencies.
Measurement Reliability	Chromium Happy Eyeballs masks failures with silent IPv4 fallback	**Decoupled Dual-Path Socket Probing:** Enforces pure IPv6 and IPv4 sockets simultaneously for true ground truth.
Failure Classification	Binary Pass/Fail or generic connection error	**4-Stage Protocol Isolation:** Pinpoints exact failure layer: DNS, TCP, TLS, or HTTP.
Dependency Visualization	Flat tabular list of hostnames	**Interactive Causal DAG:** Visual network tree showing parent-child dependencies and blast radius impact.
Performance	Serial crawl (30–60s per website)	**Async Concurrent Micro-Probing:** Evaluates 100+ dependencies in under 3 seconds using asyncio.

Solution Architecture & UML Design

Component Architecture & Diagnostic Pipeline

Layer	Component	Functionality & Output
Presentation	Web UI & Graph Visualizer (Canvas / Cytoscape DAG)	Interactive dependency graph, real-time audit progress, 30-site benchmark explorer, and CSV/JSON export.
Application API	FastAPI Auditor Backend	REST endpoints, WebSocket progress streaming, benchmark runner, and report aggregator.
Ingestion & Crawl	Headless Chromium Interceptor	Executes DOM scripts, captures request/response events, records initiators, MIME types, and sizes.
Protocol Engine	4-Stage Diagnostic Gauntlet (DNS, TCP, TLS, HTTP)	Stage 1: A/AAAA queries; Stage 2: AF_INET6 TCP SYN; Stage 3: SNI TLS handshake; Stage 4: HTTP GET & TTFB.
Ground Truth	Controlled Testbed Server	Embedded test site seeded with deliberate IPv4-only, blackholed, and TLS-mismatched assets (verifies 100% precision/recall).

Data Flow Diagram (DFD)

End-to-End Information & Pipeline Execution Flow

Step	Data Stage	Transformation & Protocol Actions
1. Input	Target URL Submission	User submits URL (e.g. https://target.in) or selects from 30 national benchmark targets.
2. Crawl	Browser Runtime Capture	Headless browser loads target; extracts resource URLs, MIME types, response headers, and caller initiator trees.
3. Graph	DAG Construction	Builds in-memory dependency tree: Apex Domain → 1st-Party Assets → 3rd-Party Providers.
4. Probe	Concurrent 4-Stage Probes	Dispatches parallel non-blocking probes across DNS (A/AAAA), TCP (IPv6 SYN), TLS (SNI), and HTTP (GET).
5. Score	Readiness & Impact Aggregation	Calculates IPv6 Readiness Score (%), classifies failure stages, and measures critical-path blast radius.
6. Output	Visual & Data Artifacts	Renders interactive colored graph, generates CSV/JSON audit reports, and logs measurement telemetry context.

Deployment Timeline & Hackathon Milestones

Phased Execution Roadmap (AIORI 3.0)

Milestone	Target Dates	Key Deliverables & Outcomes
Phase 1: Foundation (Level 1 Qualification)	Oct 11 – Oct 14, 2026	• Complete Steps 1–7: GitHub repo setup, collaborator invite, presentation PDF upload, and intro video. • Finalize core architecture, testbed design, and 30-site target selection.
Phase 2: Core Build & Mentoring (Level 2 Execution)	Oct 16 – Oct 31, 2026	• Attend scheduled AIORI mentoring sessions with prototype demo and pseudocode. • Implement headless browser interceptor, 4-stage diagnostic engine, and controlled testbed.
Phase 3: Empirical Study & Evaluation	Oct 31 – Nov 04, 2026	• Execute full 30-site empirical evaluation across Indian e-Gov, e-Commerce, news, and fintech sectors. • Generate comprehensive CSV/JSON datasets, technical reports, and working dashboard.
Phase 4: Regional Rounds & Finals	Nov 24 – Dec 17, 2026	• Regional presentation at designated center (Bengaluru / Pune / Kolkata / Noida). • Grand Finale presentation at IIT Jammu (Top 8 teams).

Bibliography & Standards References

References, RFCs & Research Foundations

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Thank You for Your Attention!

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We look forward to demonstrating our prototype to the mentors.